Building Data Lakes on AWS

Module 1
Module 1: Introduction to Data Lakes

Module 2
Module 2: Data ingestion, cataloging, and preparation

Module 3
Module 3: Building a data lake with AWS Lake Formation
Welcome to Module 3 Video • 34 sec
AWS Lake Formation Overview Video • 2 min
AWS Lake Formation Basic Permission Model Video • 5 min
Reading: AWS Lake Formation Basics Reading • 15 min
Lab 1: Building a Data Lake using AWS Lake Formation Graded App Item • 1h
Knowledge Check Graded Assignment • Grade: 100%

Module 4
Module 4: Data processing and analytics
Module Objectives Video • 50 sec
Data Transformation Video • 3 min
Data Processing with AWS Glue Video • 4 min
AWS Glue Jobs and Workflows Video • 2 min
Demo - Glue Databrew Video • 9 min
Demo - Glue Studio Video • 8 min
AWS Glue Discussion Prompt • 10 min
Tech Talk - Glue / Athena Federated Queries Video • 6 min
Knowledge Check Graded Assignment • Grade: 100%

Module 5
Module 5: AWS Lake Formation additional configurations and capabilities
Module Objectives Video • 1 min
Using Blueprints and Workflows Video • 2 min
Fine-grained Access Control Video • 5 min
Demo - LF-Tags Video • 6 min
Visualizing Data with QuickSight - Demo Video • 4 min
Lab 2: Automate Data Lake Creation Using AWS Lake Formation Blueprints Graded App Item • 1h
Knowledge Check Graded Assignment • Grade: 100%

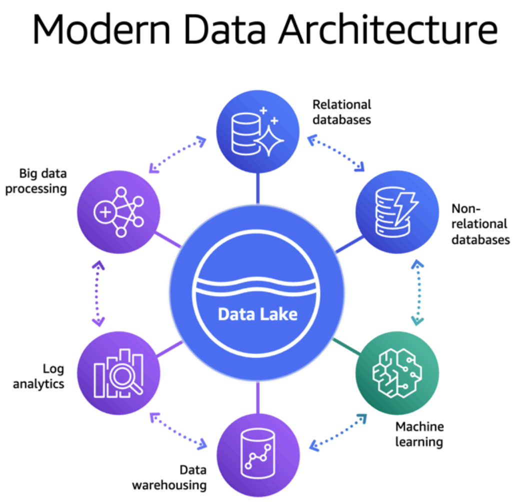
Module 6
Module 6: Modern data architecture on AWS
Module Objectives Video • 43 sec
Modern Data Architecture Video • 2 min
Reading: Modern data architecture Reading • 5 min
Data Movement Scenarios Video • 3 min
Reading: Data Movement Scenarios Reading • 15 min
Data Sharing Models Video • 3 min
Tech Talk - What's your favorite data lake feature? Video • 5 min
Lab 3: Publishing and Managing Data Product in AWS Lake Formation Graded App Item • 1h
Course Recap/Key Takeaways Video • 3 min
Post-Assessment Graded Assignment • 1h

Reading: Data Movement Scenarios

Organizations have been building data lakes to analyze massive amounts of data for deeper insights into their data. To do this, they bring data from multiple silos into their data lake, and then run analytics and AI/ML directly on it. It is also common for these organizations to have data stored in specialized data stores, such as a NoSQL database, a search service, or a data warehouse, to support different use cases. To analyze all of the data that is spread across the data lake and other data stores efficiently, businesses often move data in and out of the data lake and between these data stores. This data movement can get complex and messy as the data grows in these data stores.

To address this, businesses need a data architecture that allows building scalable, cost-effective data lakes. The architecture can also support simplified governance and data movement between various data stores. We refer to this as a modern data architecture. Modern data architecture integrates a data lake, a data warehouse, and other purpose-built data stores while enabling unified governance and seamless data movement.

As shown in the following diagram, with a modern data architecture, organizations can store their data in a data lake and use purpose-built data stores that work with the data lake. This approach allows access to all of the data to make better decisions with agility.



There are three different patterns for data movement. They can be described as follows:

Inside-out data movement

A subset of data in a data lake is sometimes moved to a data store, such as an Amazon OpenSearch Service cluster or an Amazon Neptune cluster. This pattern supports specialized analytics, such as search analytics, building knowledge graphs, or both. For example, enterprises send information from structured sources (such as relational databases), unstructured sources (such as metadata, media, or spreadsheets) and other assets to a data lake. From there, it is moved to Amazon Neptune to build a knowledge graph. We refer to this kind of data movement as inside-out.

Outside-in data movement

Organizations use data stores that best fit their applications and later move that data into a data lake for analytics. For example, to maintain game state, player data, session history, and leader boards, a gaming company might choose Amazon DynamoDB as the data store. This data can later be exported to a data lake for additional analytics to improve the gaming experience for their players. We refer to this kind of data movement as outside-in.

Around the perimeter

In addition to the two preceding patterns, there are scenarios where the data is moved from one specialized data store to another. For example, enterprises might copy customer profile data from their relational database to a NoSQL database to support their reporting dashboards. We refer to this kind of data movement as around the perimeter.

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coach

Venkatesh
Give me a summary

Coach
TheThe content focuses on the concept of modern data architecture and its role in managing data lakes effectively.

Understanding Modern Data Architecture

- Organizations build data lakes to analyze large volumes of data for deeper insights, integrating data from various silos.
- A modern data architecture combines data lakes, data warehouses, and specialized data stores, enabling unified governance and efficient data movement.

Data Movement Patterns

- **Inside-out Data Movement**: Data is moved from a data lake to specialized data stores for specific analytics, such as building knowledge graphs.
- **Outside-in Data Movement**: Data from specialized stores is moved into a data lake for broader analytics, enhancing decision-making.

Perimeter Data Movement

- Data can also be transferred between specialized data stores, such as moving customer profile data from a relational database to a NoSQL database for reporting purposes.

Ask me anything